

Activation functions

- To make work more efficient and for exact output, some force or activation is given. Like that, activation function is applied over the net input to calculate the output of an ANN.
- Information processing of processing element has two major parts: input and output. An integration function (f) is associated with input of processing element.
- Several activation functions are there.

1. Identity function:

- it is a linear function which is defined as

$$f(x) = x \text{ for all } x$$

- The output is same as the input.

2. Binary step function

- it is defined as

$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ 0 & \text{if } x < \theta \end{cases}$$

where θ represents threshold value.

It is used in single layer nets to convert the net input to an output that is binary. (0 or 1)

3. Bipolar step function:

It is defined as

$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ -1 & \text{if } x < \theta \end{cases}$$

where θ represents threshold value. Used in single layer nets to convert the net input to an output that is bipolar (+1 or -1).

4. Sigmoid function

used in Back propagation nets.

Two types:

a) binary sigmoid function

-logistic sigmoid function or unipolar sigmoid function.

-it is defined as

$$f(x) = \frac{1}{1 + e^{-\lambda x}}$$

where λ – steepness parameter.

-The derivative of this function is

$f'(x) = \lambda f(x)[1-f(x)]$. The range of sigmoid function is 0 to 1.

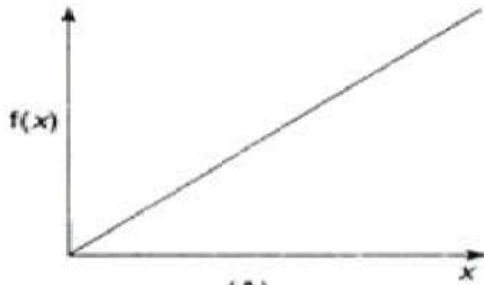
b) Bipolar sigmoid function

$$f(x) = \frac{2}{1 + e^{-\lambda x}} - 1 = \frac{1 - e^{-\lambda x}}{1 + e^{-\lambda x}}$$

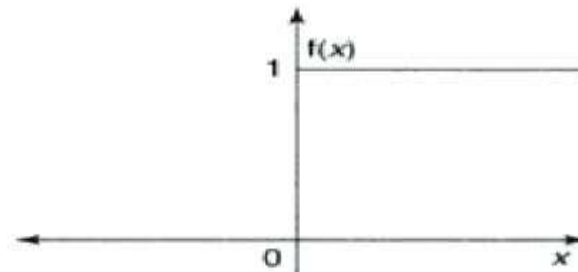
where λ - steepness parameter and the sigmoid range is between -1 and +1.

5. Ramp function

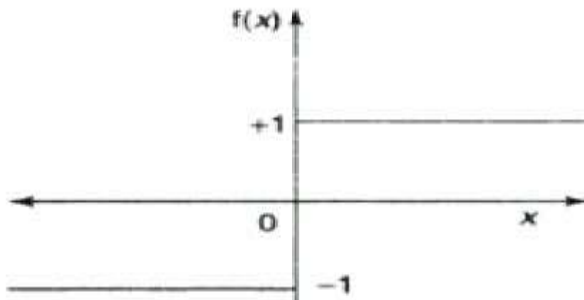
$$f(x) = \begin{cases} 1 & \text{if } x > 1 \\ x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{if } x < 0 \end{cases}$$



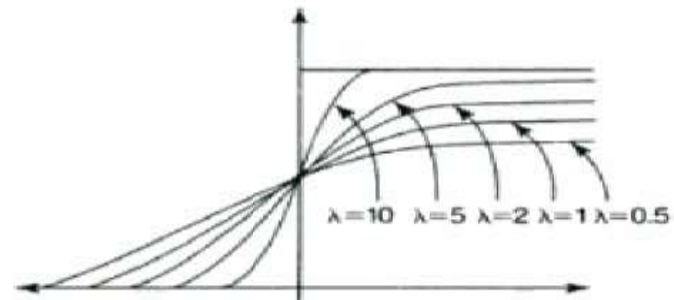
(A)



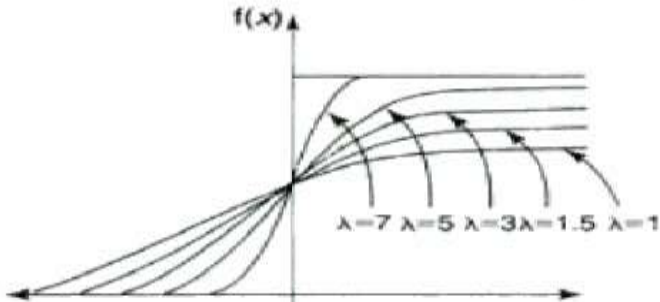
(B)



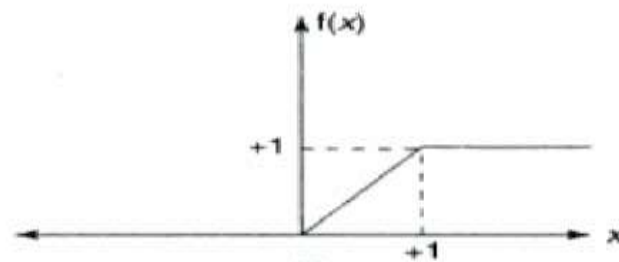
(C)



(D)



(E)



(F)

2-15 Depiction of activation functions: (A) identity function; (B) binary step function; (C) bipolar step function; (D) binary sigmoidal function; (E) bipolar sigmoidal function; (F) ramp function.